Shafizur Rahman Seeam

Ph.D. Student in Computing and Information Sciences
Department of Cybersecurity, Rochester Institute of Technology

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Research Interests

Privacy-Enhancing Technologies • Differential Privacy • AI Agent Privacy

Education

Rochester Institute of Technology, Rochester, NY *Jan. 2023 – Present*
Ph.D. in Computer and Information Sciences

- Advisor: Dr. Yidan Hu
- Dissertation: *Beyond Isolated Decisions: Advancing Dependence-Aware User-Side Privacy Mechanisms*

University of Dhaka, Dhaka, Bangladesh *Jan. 2015 – Dec. 2019*
B.Sc. in Electrical and Electronic Engineering

Publications

Under Review

[1] **Shafizur Rahman Seeam**, Zhengxiong Li, Zhiyuan Yu, Yimin Chen, and Yidan Hu. **PrivScope: Task-scoped Disclosure Control for Information-seeking Hybrid Agentic Systems.** *Manuscript under review*, 2026.

Peer-Reviewed Publications

[2] **Shafizur Rahman Seeam**, Ye Zheng, Zhengxiong Li, and Yidan Hu. **PrivAR: Client-Side Privacy Framework for Real-Time Location-Based Augmented Reality.** *46th IEEE International Conference on Distributed Computing Systems (ICDCS)*, Seoul, South Korea, 2026.

[3] **Shafizur Rahman Seeam**, Ye Zheng, and Yidan Hu. **Frequency Estimation of Correlated Multi-attribute Data under Local Differential Privacy.** *Proceedings on Privacy Enhancing Technologies (PoPETs)*, 2026.

[4] Ye Zheng, **Shafizur Rahman Seeam**, Yidan Hu, Rui Zhang, and Yanchao Zhang. **Locally Differentially Private Frequency Estimation via Joint Randomized Response.** *Proceedings on Privacy Enhancing Technologies (PoPETs)*, 2025.

[5] **Shafizur Rahman Seeam**, Zhengxiong Li, and Yidan Hu. **A Black-Box Approach for Quantifying Leakage of Trace-Based Correlated Data.** *7th International Workshop on Physics Embedded AI Solutions in Mobile Computing (PICASSO), ACM MobiCom*, 2024.

[6] Swapnil Sayan Saha, **Shafizur Rahman Seeam**, Mosabber Uddin Ahmed, and Subrata Kumar Aditya. **Ensuring Cybersecure Telemetry and Telecommand in Small Satellites: Recent Trends and Empirical Propositions.** *IEEE Aerospace and Electronic Systems Magazine*, vol. 34, no. 8, pp. 34–49, 2019.

[7] Swapnil Sayan Saha, **Shafizur Rahman Seeam**, Zarin Rezwana Ridita Haque, Tahera Hossain, Sozo Inoue, and Md Atiqur Rahman Ahad. **Position Independent Activity Recognition Using Shallow Neural Architecture and Empirical Modeling.** *ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC)*, 2019.

[8] Swapnil Sayan Saha, **Shafizur Rahman Seeam**, Miftahul Jannat Rasna, Tahera Hossain, Sozo Inoue, and Md Atiqur Rahman Ahad. **Supervised and Neural Classifiers for Locomotion Analysis.** *ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp)*, pp. 1563–1570, 2018.

[9] Swapnil Sayan Saha, **Shafizur Rahman Seeam**, Miftahul Jannat Rasna, Tarek Bin Zahid, A.K.M. Mahfuzul Islam, and Md Atiqur Rahman Ahad. **Feature Extraction, Performance Analysis, and System Design Using the DU Mobility Dataset.** *IEEE Access*, vol. 6, pp. 44776–44786, 2018.

[10] Swapnil Sayan Saha, **Shafizur Rahman Seeam**, Miftahul Jannat Rasna, A.K.M. Mahfuzul Islam, and Md Atiqur Rahman Ahad. **DU-MD: An Open-Source Human Action Dataset for Ubiquitous Wearable Sensors.** *7th International Conference on Informatics, Electronics & Vision (ICIEV)*, Kitakyushu, Japan, pp. 567–572, 2018. **Best Paper Award Candidate**.

Research Experience

Graduate Research Assistant — Rochester Institute of Technology *Jan. 2023 – Present*
Advisor: Dr. Yidan Hu

- **Privacy-preserving agentic systems:** Develop user-side disclosure-control mechanisms for hybrid local–cloud agents, including task-scoped control over what cloud-bound information is disclosed and at what specificity.
- **Differential privacy:** Design dependence-aware mechanisms for multi-attribute frequency estimation under local differential privacy, improving utility while preserving formal privacy guarantees.
- **Location privacy:** Develop client-side privacy mechanisms for continuous, latency-sensitive location reporting in real-time location-based augmented-reality applications.

Teaching Experience

Instructor (Upcoming) — Authentication and Security Models (CSEC 472) *Spring 2027*
Rochester Institute of Technology, Rochester, NY

- Appointed as the primary instructor for approximately 35 students, with full responsibility for course planning and delivery, lectures, laboratory and assignment development, project supervision, student mentoring, and assessment.

Graduate Teaching Assistant *Aug. 2024 – Present*
Rochester Institute of Technology, Rochester, NY

- *Advanced Data Privacy (CSEC 721)* — Spring 2026; contributed to course and project design, graded student work, and mentored students throughout their projects.
- *Authentication and Security Models (CSEC 472)* — Fall 2024, Spring 2025, Fall 2025, and Fall 2026; contributed to course design, created and graded laboratory assignments, and evaluated course projects and other submissions.
- *Graduate Seminar in Computing Security (CSEC 759)* — Spring 2025.

Professional Experience

Co-Founder & Director — Orboroi AB, Stockholm, Sweden *Jul. 2021 – Present*

- Lead Bangladesh operations, including team coordination, client engagement, and project delivery.
- Oversee financial operations, including cash flow, payments, and expense control.
- Deploy and maintain the company website, database, CVAT platform, and Linux servers.

Awards & Honors

- **PETS 2026 Student Travel Stipend Award**, 2026.
- **Best Paper Award Candidate**, ICIEV, Kitakyushu, Japan, 2018.
- **3rd Place**, Digital Khichuri Challenge, UNDP & ICT Division, Bangladesh, 2017.
- **1st and 2nd Place**, Bangladesh Electronics Olympiad Project Competition, 2015 and 2017.
- **Champion, Project Showcasing (Junior)**, EEE Day, BUET, 2015 and 2016.
- **Runner-Up, Project Showcasing**, Mecceleration, IUT, 2015.

Academic & Professional Service

- Judge, GENIUS Olympiad international high-school project competition, 2023 and 2024.

Technical Skills

Programming & AI: Python, Kotlin, SQL, MATLAB, TensorFlow, LangChain, Ollama
Systems & Tools: Linux, Node.js, MongoDB, Android, CVAT, Git, LaTeX